

The planning algorithm generates a function $\mathbf{q}(t)$ interpolating the given vectors of joint variables at each point, in respect of the imposed constraints.

In general, a joint space trajectory planning algorithm is required to have the following features:

- the generated trajectories should be not very demanding from a computational viewpoint,
- the joint positions and velocities should be continuous functions of time (continuity of accelerations may be imposed, too),
- undesirable effects should be minimized, e.g., nonsmooth trajectories interpolating a sequence of points on a path.

At first, the case is examined when only the initial and final points on the path and the traveling time are specified (*point-to-point*); the results are then generalized to the case when also intermediate points along the path are specified (*motion through a sequence of points*). Without loss of generality, the single joint variable $q(t)$ is considered.

4.2.1 Point-to-Point Motion

In *point-to-point motion*, the manipulator has to move from an initial to a final joint configuration in a given time t_f . In this case, the actual end-effector path is of no concern. The algorithm should generate a trajectory which, in respect to the above general requirements, is also capable of optimizing some performance index when the joint is moved from one position to another.

A suggestion for choosing the motion primitive may stem from the analysis of an incremental motion problem. Let I be the moment of inertia of a rigid body about its rotation axis. It is required to take the angle q from an initial value q_i to a final value q_f in a time t_f . It is obvious that infinite solutions exist to this problem. Assumed that rotation is executed through a torque τ supplied by a motor, a solution can be found which minimizes the energy dissipated in the motor. This optimization problem can be formalized as follows. Having set $\dot{q} = \omega$, determine the solution to the differential equation

$$I\dot{\omega} = \tau$$

subject to the condition

$$\int_0^{t_f} \omega(t) dt = q_f - q_i$$

so as to minimize the performance index

$$\int_0^{t_f} \tau^2(t) dt.$$

It can be shown that the resulting solution is of the type

$$\omega(t) = at^2 + bt + c.$$

Even though the joint dynamics cannot be described in the above simple manner,¹ the choice of a third-order polynomial function to generate a joint trajectory represents a valid solution for the problem at issue.

Therefore, to determine a joint motion, the *cubic polynomial*

$$q(t) = a_3t^3 + a_2t^2 + a_1t + a_0 \quad (4.1)$$

can be chosen, resulting into a parabolic velocity profile

$$\dot{q}(t) = 3a_3t^2 + 2a_2t + a_1$$

and a linear acceleration profile

$$\ddot{q}(t) = 6a_3t + 2a_2.$$

Since four coefficients are available, it is possible to impose, besides the initial and final joint position values q_i and q_f , also the initial and final joint velocity values $\dot{q}_i$ and $\dot{q}_f$ which are usually set to zero. Determination of a specific trajectory is given by the solution to the following system of equations:

$$\begin{aligned} a_0 &= q_i \\ a_1 &= \dot{q}_i \\ a_3t_f^3 + a_2t_f^2 + a_1t_f + a_0 &= q_f \\ 3a_3t_f^2 + 2a_2t_f + a_1 &= \dot{q}_f, \end{aligned}$$

that allows the computation of the coefficients of the polynomial in (4.1).² Figure 4.1 illustrates the timing law obtained with the following data: $q_i = 0$, $q_f = \pi$, $t_f = 1$, and $\dot{q}_i = \dot{q}_f = 0$. As anticipated, velocity has a parabolic profile, while acceleration has a linear profile with initial and final discontinuity.

If it is desired to assign also the initial and final values of acceleration, six constraints have to be satisfied and then a polynomial of at least *fifth* order is needed. The motion timing law for the generic joint is then given by

$$q(t) = a_5t^5 + a_4t^4 + a_3t^3 + a_2t^2 + a_1t + a_0, \quad (4.2)$$

whose coefficients can be computed, as for the previous case, by imposing the conditions for $t = 0$ and $t = t_f$ on the joint variable $q(t)$ and on its first two derivatives. With the choice (4.2), one obviously gives up minimizing the above performance index.

An alternative approach with timing laws of blended polynomial type is frequently adopted in industrial practice, which allows a direct verification

¹ In fact, recall that the moment of inertia about the joint axis is a function of manipulator configuration.

² Notice that it is possible to normalize the computation of the coefficients, so as to be independent both on the final time t_f and on the path length $|q_f - q_i|$.

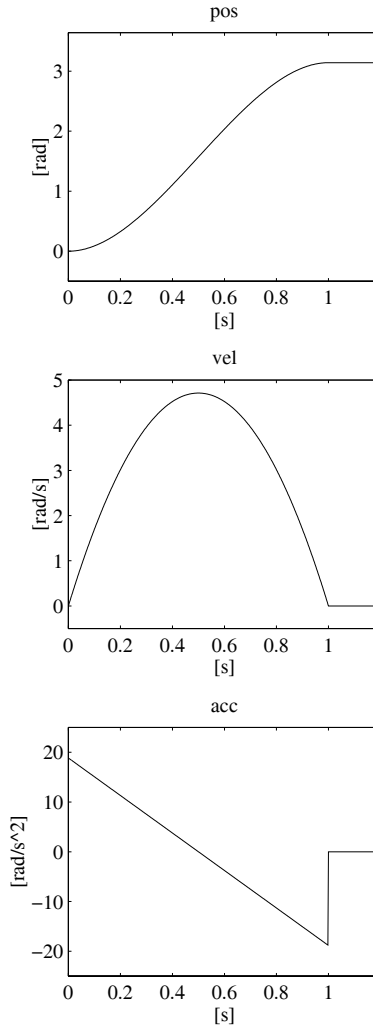


Fig. 4.1. Time history of position, velocity and acceleration with a cubic polynomial timing law

of whether the resulting velocities and accelerations can be supported by the physical mechanical manipulator.

In this case, a *trapezoidal velocity profile* is assigned, which imposes a constant acceleration in the start phase, a cruise velocity, and a constant deceleration in the arrival phase. The resulting trajectory is formed by a linear segment connected by two parabolic segments to the initial and final positions.

In the following, the problem is formulated by assuming that the final time of trajectory duration has been assigned. However, in industrial practice, the user is offered the option to specify the velocity percentage with respect to the maximum allowable velocity; this choice is aimed at avoiding occurrences when

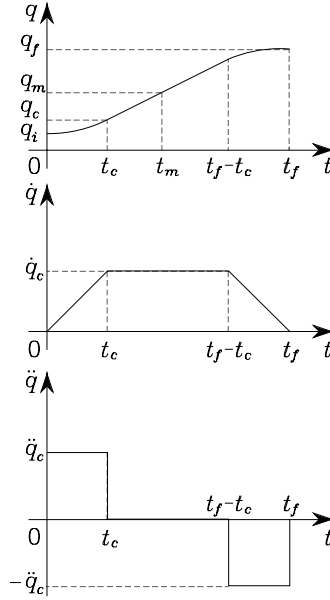


Fig. 4.2. Characterization of a timing law with trapezoidal velocity profile in terms of position, velocity and acceleration

the specification of a much too short motion duration would involve much too large values of velocities and/or accelerations, beyond those achievable by the manipulator.

As can be seen from the velocity profiles in Fig. 4.2, it is assumed that both initial and final velocities are null and the segments with constant accelerations have the same time duration; this implies an equal magnitude $\ddot{q}_c$ in the two segments. Notice also that the above choice leads to a symmetric trajectory with respect to the average point $q_m = (q_f + q_i)/2$ at $t_m = t_f/2$.

The trajectory has to satisfy some constraints to ensure the transition from q_i to q_f in a time t_f . The velocity at the end of the parabolic segment must be equal to the (constant) velocity of the linear segment, i.e.,

$$\ddot{q}_c t_c = \frac{q_m - q_c}{t_m - t_c} \quad (4.3)$$

where q_c is the value attained by the joint variable at the end of the parabolic segment at time t_c with constant acceleration $\ddot{q}_c$ (recall that $\dot{q}(0) = 0$). It is then

$$q_c = q_i + \frac{1}{2} \ddot{q}_c t_c^2. \quad (4.4)$$

Combining (4.3), (4.4) gives

$$\ddot{q}_c t_c^2 - \ddot{q}_c t_f t_c + q_f - q_i = 0. \quad (4.5)$$

Usually, $\ddot{q}_c$ is specified with the constraint that $\text{sgn } \ddot{q}_c = \text{sgn } (q_f - q_i)$; hence, for given t_f , q_i and q_f , the solution for t_c is computed from (4.5) as ($t_c \leq t_f/2$)

$$t_c = \frac{t_f}{2} - \frac{1}{2} \sqrt{\frac{t_f^2 \ddot{q}_c - 4(q_f - q_i)}{\ddot{q}_c}}. \quad (4.6)$$

Acceleration is then subject to the constraint

$$|\ddot{q}_c| \geq \frac{4|q_f - q_i|}{t_f^2}. \quad (4.7)$$

When the acceleration $\ddot{q}_c$ is chosen so as to satisfy (4.7) with the equality sign, the resulting trajectory does not feature the constant velocity segment any more and has only the acceleration and deceleration segments (*triangular profile*).

Given q_i , q_f and t_f , and thus also an average transition velocity, the constraint in (4.7) allows the imposition of a value of acceleration consistent with the trajectory. Then, t_c is computed from (4.6), and the following sequence of polynomials is generated:

$$q(t) = \begin{cases} q_i + \frac{1}{2} \ddot{q}_c t^2 & 0 \leq t \leq t_c \\ q_i + \ddot{q}_c t_c (t - t_c/2) & t_c < t \leq t_f - t_c \\ q_f - \frac{1}{2} \ddot{q}_c (t_f - t)^2 & t_f - t_c < t \leq t_f. \end{cases} \quad (4.8)$$

Figure 4.3 illustrates a representation of the motion timing law obtained by imposing the data: $q_i = 0$, $q_f = \pi$, $t_f = 1$, and $|\ddot{q}_c| = 6\pi$.

Specifying acceleration in the parabolic segment is not the only way to determine trajectories with trapezoidal velocity profile. Besides q_i , q_f and t_f , one can specify also the cruise velocity $\dot{q}_c$ which is subject to the constraint

$$\frac{|q_f - q_i|}{t_f} < |\dot{q}_c| \leq \frac{2|q_f - q_i|}{t_f}. \quad (4.9)$$

By recognizing that $\dot{q}_c = \ddot{q}_c t_c$, (4.5) allows the computation of t_c as

$$t_c = \frac{q_i - q_f + \dot{q}_c t_f}{\dot{q}_c}, \quad (4.10)$$

and thus the resulting acceleration is

$$\ddot{q}_c = \frac{\dot{q}_c^2}{q_i - q_f + \dot{q}_c t_f}. \quad (4.11)$$

The computed values of t_c and $\ddot{q}_c$ as in (4.10), (4.11) allow the generation of the sequence of polynomials expressed by (4.8).

The adoption of a trapezoidal velocity profile results in a worse performance index compared to the cubic polynomial. The decrease is, however, limited; the term $\int_0^{t_f} \tau^2 dt$ increases by 12.5% with respect to the optimal case.

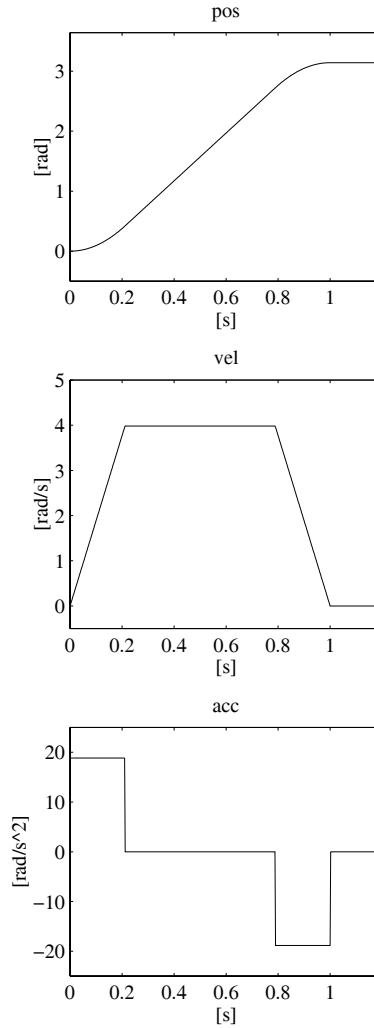


Fig. 4.3. Time history of position, velocity and acceleration with a trapezoidal velocity profile timing law

4.2.2 Motion Through a Sequence of Points

In several applications, the path is described in terms of a number of points greater than two. For instance, even for the simple point-to-point motion of a pick-and-place task, it may be worth assigning two intermediate points between the initial point and the final point; suitable positions can be set for lifting off and setting down the object, so that reduced velocities are obtained with respect to direct transfer of the object. For more complex applications, it may be convenient to assign a *sequence of points* so as to guarantee better monitoring on the executed trajectories; the points are to be specified more densely in those segments of the path where obstacles have to be avoided